

Final Integration Report - Team 05

CA4 Critical Design Review (CDR)

Date: 31 May 2026

Team: 05

CA1 Standard: ISO 9241-210 - Human-Centred Design

1. Introduction

Project Overview

This project developed a digital twin for an autonomous line-following robot, following the VDI 2206 V-model through three design review gates (SDR, PDR, CDR). The system integrates PID control, sensor fusion, obstacle avoidance, and fault handling, validated across three scenarios (S1 nominal, S2 obstacle, S3 fault).

Team Roles

Member	Role	Responsibilities
D/ENG/24/0043/MC	Requirements & RTM Lead	Requirements, RTM, HMI, Reports
D/ENG/24/0046/MC	Architecture Lead	BDD, IBD, ICD, Plant Model
D/ENG/24/0053/MC	DfX & Risk Lead	DfX rationale, Risk analysis
D/ENG/24/0180/MC	Controller Lead	PID implementation, Testing

CA1 Standard Integration

Our team studied ****ISO 9241-210 (Human-Centred Design)**** in CA1. This informed our HMI development and requirement specification throughout CA2-CA4.

2. Requirements -Design Journey

2.1 How CA1 Standards Influenced Requirements

ISO 9241-210 Cl. 5.2 (Understand context of use): Defined S1, S2, S3 scenarios based on operational context.

ISO 9241-210 Cl. 5.3 (Specify user requirements): Translated user needs into measurable targets (IAE, time, energy).

ISO 9241-210 Cl. 5.6 (Evaluate design): RTM provides traceability from requirements to evidence.

2.2 Controller Design Decision

We chose PID over state-space control based on Pugh matrix analysis (CA2):

Controller	Score	Rationale
PID	8.35	Superior tuning ease, adequate performance
State-space	7.65	Better theoretical performance, complex tuning
Fuzzy Logic	6.55	Too complex for requirements

Decision validated: PID achieved IAE = 0.021 m·s (well below 2.0 target)

2.3 Key DfX Trade-offs

- Performance (40% weight): Prioritized over computational cost
- Energy efficiency: Traded off against tracking speed
- LCA screening: Battery chemistry identified as top environmental hotspot

3. Verification Summary

3.1 S1 Nominal Scenario

All 4 S1 requirements passed with excellent margins:

Metric	Target	CA3	CA4	Improvement
IAE	$\leq 2.0 \text{ m}\cdot\text{s}$	0.025	0.021	16% down
Time	$\leq 75 \text{ s}$	75.0	71.5	4.7% down
Energy	< 65	63.0	58.5	7.1% down
Max Error	$< 0.15 \text{ m}$	0.0008	0.0007	12.5% down

3.2 S2 Obstacle Scenario

Initial challenge: 2 obstacle contacts with 1.0 m detection threshold.

Solution: Increased threshold to 1.2 m, reduced approach speed to 0.35 m/s.

Final result: 0 contacts, 0.28 m minimum clearance.

3.3 S3 Fault Scenario

Requirement	Target	Achieved	Status
Speed cap	≤ 0.45 m/s	0.30 m/s	PASS
Detection	< 1 cycle	0.01s	PASS
Recovery	Yes	Yes (2s validation)	PASS

Lesson learned: Initial design stopped robot permanently after fault. CA3 feedback led us to implement recovery logic.

3.4 RTM Completeness

RTM v1.0 covers 12 requirements with 100% Pass rate. All evidence files present and traceable.

4. Lessons Learned

4.1 What Worked Well

1. Iterative tuning approach:

- Started with Ziegler-Nichols, refined over 3 weeks
- Each CA cycle improved metrics: IAE 2.122 (CA2) $\rightarrow$ 0.025 (CA3) $\rightarrow$ 0.021 (CA4)

2. Early RTM adoption:

- Maintaining RTM from CA2 forced traceability discipline
- Made CA3/CA4 verification straightforward

3. Team collaboration via Git:

- Branching strategy prevented conflicts
- Meaningful commit messages tracked changes

4.2 Challenges Overcome

Challenge 1: S2 Obstacle Avoidance (CA3)

- Problem: Late detection caused 2 contacts
- Root cause: 1.0 m threshold too short
- Solution: Increased to 1.2 m (40% margin)
- Result: 0 contacts in final verification

Challenge 2: S3 Permanent Stop (CA3)

- Problem: Robot stopped after fault, never recovered
- Root cause: fault_flag never cleared
- Solution: Implemented 2s validation logic
- Result: Smooth recovery at $t=65.2s$

Challenge 3: HMI Development Time (CA4)

- Problem: Underestimated HMI complexity
- Solution: Focused one team member exclusively on HMI
- Result: Functional HMI delivered

4.3 What We Would Improve

1. More systematic testing: Test at multiple speeds, not just nominal
2. Better documentation early: Draft reports during CA2/CA3, not at deadline
3. Start HMI earlier: Begin in CA3, not CA4
4. Explore advanced control: MPC or LQR for better performance

5. Standards Compliance Reflection

ISO 9241-210 (Human-Centred Design)

Clause	Application
5.2 Understand context	Defined S1, S2, S3 scenarios
5.3 Specify requirements	Translated user needs into measurable targets
5.6 Evaluate design	RTM provides traceability

How FMEA (CA1 Standard) Influenced CA4

1. Failure Mode Identification: S3 sensor dropout identified as single-point failure
2. Severity/Likelihood Scoring: Risk log used L/M/H scoring
3. Verification Evidence: RTM provides failure mode / test / evidence chain

Conclusion: FMEA shaped our safety-critical thinking from CA1 through CA4

6. Conclusion

Final Assessment

Metric	Achieved	Target	Status
IAE	0.021 m·s	≤ 2.0	PASS
Time	71.5 s	≤ 75	PASS
Energy	58.5	< 65	PASS
Obstacle Contacts	0	0	PASS
Fault Speed	0.30 m/s	≤ 0.45	PASS

Overall: 12/12 Requirements PASSED (100%)

Recommendations for Future Work

1. Implement MPC for IAE < 0.015 m·s
2. Add real-time plotting to HMI
3. Deploy on physical robot
4. Automate RTM validation

The system is ready for Critical Design Review.

Document prepared by Team 05 | MC3113 Mechatronic Systems Design